

Zephyr RTOS Setup Guide (Windows)

This guide explains how to:

1. Install Zephyr RTOS from scratch
 2. Install QEMU for Windows
 3. Build the Zephyr Blinky sample
 4. Run Blinky on QEMU
 5. Build and flash Blinky on real hardware
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Prerequisites

Before starting, ensure you have:

- Windows 10 or Windows 11
- Internet connection
- Administrator privileges for software installation
- Git installed

Git Download:

<https://git-scm.com/download/win>

1. Install Zephyr RTOS

The Zephyr Project provides a comprehensive and continuously updated Getting Started Guide. Always follow the official documentation for the latest dependency versions and installation requirements.

Official Zephyr Getting Started Guide

https://docs.zephyrproject.org/latest/develop/getting_started/index.html

The guide covers:

- Installing Python
- Installing CMake
- Installing Ninja
- Installing Device Tree Compiler (DTC)
- Installing West
- Installing Zephyr SDK
- Creating a Zephyr workspace
- Downloading all Zephyr modules

Follow the **Windows** instructions from the official guide.

2. Verify Zephyr Installation

After completing the official setup, open a Zephyr Command Prompt, PowerShell, or terminal.

Verify West installation:

```
west --version
```

Expected output:

```
West version: x.x.x
```

Verify your workspace:

```
cd zephyrproject  
west list
```

You should see Zephyr and its modules listed.

3. Install QEMU on Windows

QEMU allows Zephyr applications to run on an emulator without requiring physical hardware.

Official QEMU Downloads

<https://www.qemu.org/download/>

Option 1: Windows Installer (Recommended)

Download the latest Windows installer from:

<https://qemu.eu/>

Installation Steps

1. Download the latest 64-bit installer.
2. Run the installer.
3. Install QEMU to:

```
C:\Program Files\qemu
```

4. Add the installation directory to your Windows PATH.

Example:

```
C:\Program Files\qemu
```

5. Open a new Command Prompt.

Verify the installation:

```
qemu-system-x86_64 --version
```

Expected output:

```
QEMU emulator version x.x.x
```

If a version is displayed, QEMU is installed correctly.

4. Build the Blinky Sample

The Blinky sample is Zephyr's "Hello World" application.

Navigate to the Zephyr repository:

```
cd zephyrproject\zephyr
```

Build for the QEMU x86 target:

```
west build -b qemu_x86 samples/basic/blinky
```

Successful output will contain:

```
Build files have been written to:  
build
```

5. Run Blinky on QEMU

After a successful build, launch the emulator:

```
west build -t run
```

A QEMU window will open.

You should see output similar to:

```
*** Booting Zephyr OS ***  
LED state: ON  
LED state: OFF  
LED state: ON  
LED state: OFF
```